Mean-band trading algorithm (`mean_band`)

This document summarizes how **forecast rows** are produced and how each **trading tick** runs stop-loss, **mean-band** TA (`strategy_id`: `mean_band`, `MeanBandTechnicalAnalysis` in `fxagent.trading.mean_band_ta`), and the venue **executor**. It stays descriptive — not a prompt playbook.

1. System architecture: forecasting vs trading tick

Two pipelines cooperate:

1. **Forecasting** builds `PredictorResultModel` rows: optional **trigger gate** (RSS-only today), **research** sentiment fetch, **predictor** LLM call, then DB persistence. Live paths often **enqueue** a worker message after a **new** row.
2. **Trading tick** (`run_trade_evaluation` → `run_trading_pipeline` in `fxagent.trading.pipeline`) loads existing rows with **`create_if_missing=False`** — it does **not** re-run the predictor — builds `PreparedTradeEvaluation`, runs **stop-loss first**, then TA, then the executor.

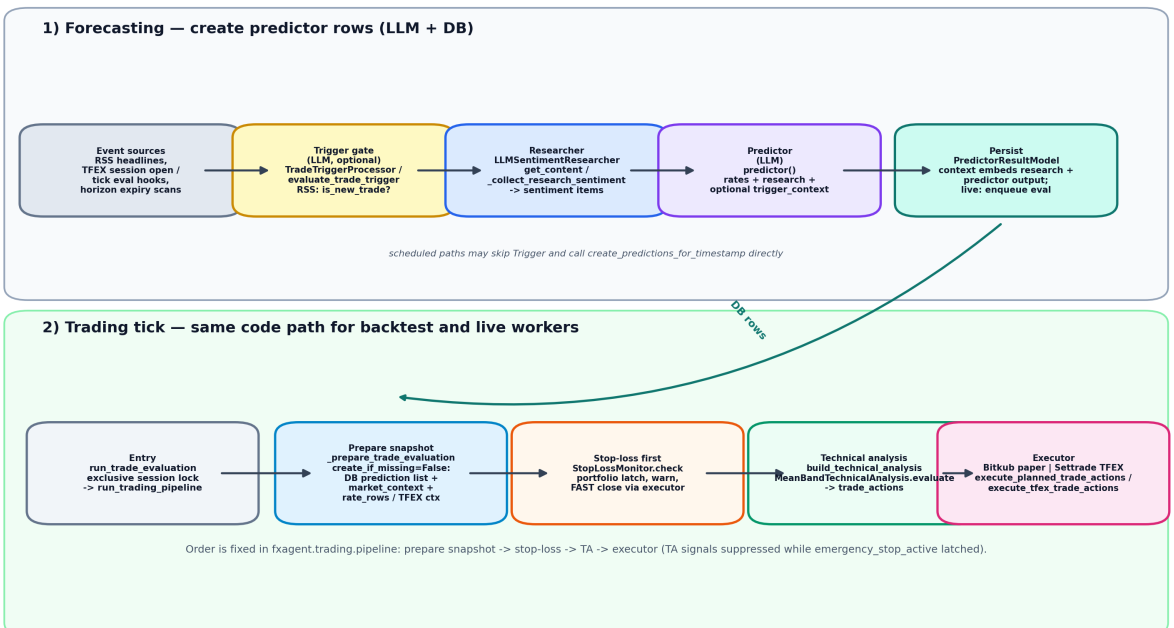

Figure 1: Forecasting chain (trigger / researcher / predictor) and trading tick (prepare, stop-loss, TA, executor).

RSS-shaped example: Google RSS → `TradeTriggerProcessor` / `evaluate_trade_trigger` → when `is_new_trade`, `TradingSession.create_prediction_with_trigger` → `create_predictions_for_timestamp` → `_collect_research_sentiment` → `_get_or_create_prediction` → `predictor()` → persist → (live) enqueue eval.

Scheduled example: horizon expiry / TFEX pre-open scanners call `create_predictions_for_timestamp` directly — trigger LLM skipped.

Core terminology

- **Base currency (X)**: e.g. USD (`USA` ISO) — first in pair notation **X/Y**
- **Quote currency (Y)**: e.g. THB (`THA`) — second in **X/Y**
- **Rate**: **Y per X** (how much quote you pay for one unit of base), e.g. THB per USD for USD/THB
- If the **rate goes up**, **X strengthens** vs Y (base appreciates); equivalently Y weakens vs X (quote depreciates)
- If the **rate goes down**, **X weakens** vs Y (base depreciates); equivalently Y strengthens vs X (quote appreciates)

All `*_appreciation` / `*_depreciation` prediction labels refer to the **base X** ("does the first currency in the pair strengthen or weaken?").

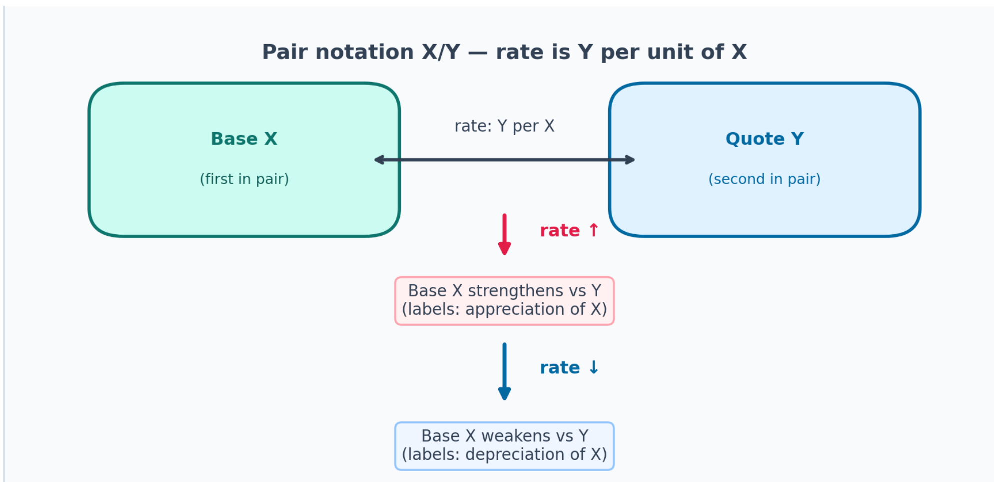

Figure 2: Illustration: currency pair X/Y and how moves in the quoted rate relate to base strength.

Examples (THB per USD):

- `32 → 33`: USD **appreciates** vs THB → label family `*_appreciation`
- `33 → 32`: USD **depreciates** vs THB → label family `*_depreciation`

Prediction inputs

Directional labels (must pair with a valid horizon when persisted):

- `high_appreciation`, `medium_appreciation`, `low_appreciation`
- `high_depreciation`, `medium_depreciation`, `low_depreciation`
- `not_confident` (no directional trade sizing from the predictor tier map)

Persisting forecasts (`predictor_result`)

- Each **directional** row needs a whole-number `time_range_hours` from the model. Missing or invalid horizons are **logged only** — nothing is written for that attempt.
- Stored horizons are **clamped** to **[1, 168]** hours (one week ceiling) before persistence.
- `predicted_ts` is **forecast horizon end** = `created_at + time_range` (wall-clock from the run, not "calendar midnight + hours").
- `asset_a` = base X ISO, `asset_b` = quote Y ISO.

- Session metadata includes `context.prediction_session_date` (`YYYY-MM-DD`) for joins and caching.

Which row mean-band TA uses

Predictions loaded for a leg are ordered **newest first**, limited to rows **active at `market_as_of`** (`created_at <= as-of <= predicted_ts`), with a **stale fallback** to older rows when none are active.

The TA picks **one effective row** per leg (`_ta_prediction_row_for_leg`):

1. If the **head** row is directional → use it.
2. If the head is `not_confident` and **still inside its horizon** → scan newer→older siblings for the **first directional** row still in play.
3. If the head is `not_confident` and **past its horizon** → **no row** (TA skips the leg — stale pause).
4. If every candidate in the active window is `not_confident` → **no row** (TA does nothing for that leg).

So **`not_confident` does not always idle the strategy**: an older directional forecast can still arm bands until its horizon ends.

Sizing (`invest_pct_from_predictions`) reads the **head** list element's text; a `not_confident` head yields **0%** deployment even if TA later resolved a directional sibling for geometry.

Market window and price sources

The SMA uses **closes** over a timedelta `range_delta` chosen in order:

1. Session `fixed_time_range_hours` (when set) — overrides predictor horizons for the SMA window only.
2. Else `time_range` on the effective predictor row (when valid).
3. Else the latest **older** row for the same leg/models with a non-null `time_range`.
4. Else **1 day**.

Spot / Bitkub: closes from `get_bitkub_closes_in_range` for the mapped symbol (when present).

TFEX (Settrade): closes from `get_tfex_mean_band_closes_with_bitkub_basis_proxy` — TFEX marks aligned with Bitkub basis where the venue series has gaps; live marks may splice in when caches are wired (`mean_band_ta._resolve_tfex_mean_band_closes`).

Bands are evaluated against **executable quotes** from market context: `buy_rate` (ask side) vs the lower edge, `sell_rate` (bid side) vs the upper edge, falling back to the latest close when the book is absent.

Band mathematics

Given closes in the window:

- `SMA` = arithmetic mean of closes.
- `std` = **population** standard deviation (`ddof=0`).
- **Half-offset** = `max(k * std, SMA_BAND_MIN_HALF_WIDTH)` with `SMA_BAND_MIN_HALF_WIDTH = 0.01` (rate units). This prevents the band collapsing when variance is tiny or zero.
- `top = SMA + half-offset`, `bottom = SMA - half-offset`.

Session `k` defaults to `0.5` (`TradeConfig.k`).

Interpretation:

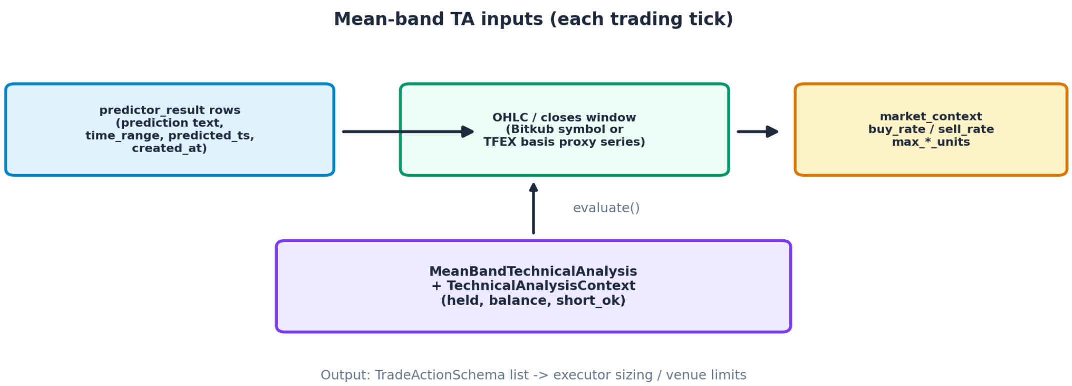

Figure 3: Inputs combined inside MeanBandTechnicalAnalysis each eval (closes + predictor rows + book context).

- `bottom` — lower edge; **buy / cover** geometry arms when `buy_rate <= bottom` (after TFEX tick checks below).
- `top` — upper edge; **sell / short** geometry arms when `sell_rate >= top` (after TFEX tick checks).

Label gates (substring match on prediction text)

`mean_band_actions_for_leg` lower-cases the effective row's `prediction` string and tests substrings `appreciation` / `depreciation`. Venue flag `require_label_gates` is `True` on **Bitkub spot** (no `settrade_symbol`) and `False` on **TFEX** sessions — this only changes **risk closes** (cover / close long), not **new opens**.

Mean-band leg	Geometry (plus TFEX 2dp tick gate when applicable)	**New open** label rule	**Close / cover** label rule
Open long from flat (`held ==` 0, balance > 0)	`buy_rate <= bottom`	Must contain `appreciation` (spot **and** TFEX)	—
Cover short (`held <` 0)	`buy_rate <= bottom`	—	**Spot:** must contain `appreciation`. **TFEX: not** gated by label — band geometry only.
Close long (`held >` 0)	`sell_rate >= top`	—	**Spot:** must contain `depreciation`. **TFEX: not** gated — geometry only.

Mean-band leg	Geometry (plus TFEX 2dp tick gate when applicable)	**New open** label rule	**Close / cover** label rule
Open short (TFEX; `held >= 0`, `short_ok`)	`sell_rate >= top`	Must contain **`depreciation`**	—

Already net long: the lower band never **adds** long size — only **opens from flat** or **covers shorts**. Pyramiding waits for an upper-band exit or regime pass.

TFEX reminder: opens still **always** require the matching directional substring; derivatives mode only relaxes label checks on **flattening** legs so positions are not trapped by a stale headline.

TFEX-only mechanics (derivatives sessions)

When **`session.settrade_symbol`** is set:

0.01 tick co-linearity gate (`apply_settrade_tick_gating`)

Before arming lower or upper-band legs, quotes and bands are snapped to **two decimal places**. If **`buy_rate`**, **`bottom`**, and **`SMA`** (lower side) or **`sell_rate`**, **`top`**, and **`SMA`** (upper side) collapse to the **same** tick, the signal **does not emit** until a distinct tick separates them — avoids duplicate ticks / fragile limits.

Regime tracking and flatten-on-flip

The session stores **`_tfex_last_prediction_regime`** (`appreciation` / `depreciation` from substring match). When the regime **flips**, a **flatten** leg may fire first (close long on flip toward depreciation; cover short on flip toward appreciation) so the opposite side can arm on a later pass.

- **Running tier score** uses the same row filter as force-close math: **Bangkok TFEX segment** bounds for `created_at`.
- **Sign confirmation:** a flatten requires the **summed tier score** to be **`> 0`** when entering **`appreciation`** and **`< 0`** when entering **`depreciation`** (clamped total in **+/-2**).
- If the score is `None` (e.g. only `not_confident` rows in the window), the flip is **deferred** — treated as lacking confirmation.
- An additional guard **blocks** some flips when the working quote sits exactly on the **SMA tick** in the adverse direction (long holder flipping out on depreciation vs bid-at-SMA symmetry — see `MeanBandTechnicalAnalysis.evaluate`).

Force-close vs forecast disagreement

If still **short** under an **`appreciation`** regime (or **long** under **`depreciation`**) and **no** covering band leg fired this pass, the TA may **force flat** — but **only** when the running tier score confirms magnitude (**`>= 0.5`** appreciation side for shorts, **`<= -0.5`** depreciation side for longs). If the score is `None`, force-close **does not** fire.

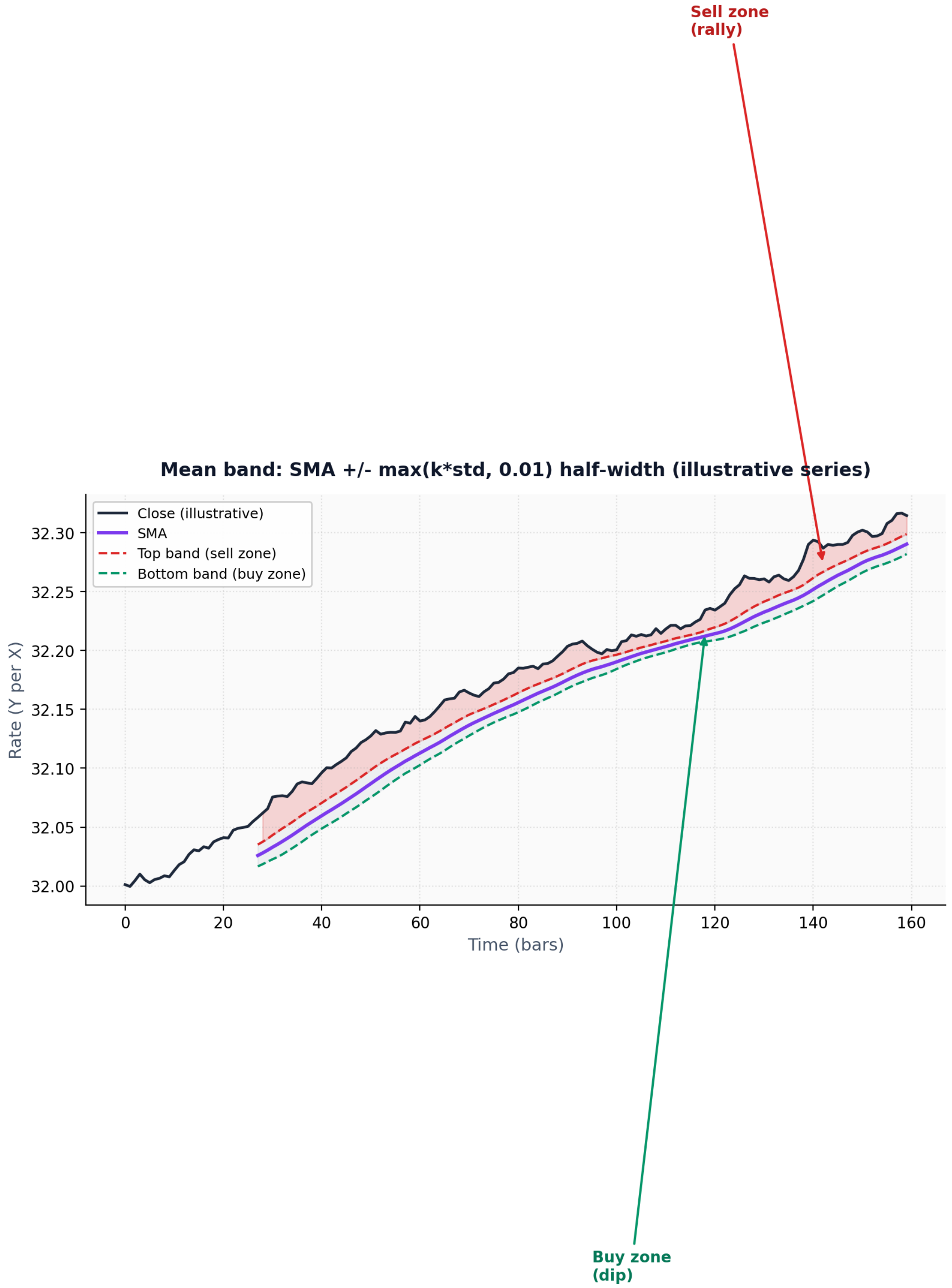

Figure 4: Illustration: rolling mean with +/- k sigma bands; preferred buy zone near the lower band and sell zone near the upper band.

Tier magnitudes (running sum)

Per directional row (excluding `not_confident`): **`high_* -> 1`**, **`medium_*`** / **`med_* -> 0.5`**, **`low_* -> 0.25`**; other directional text without a tier falls back to **0.5**. Signs follow appreciation (+) vs depreciation (-).

Spot note on the same tier-sum machinery

The **same** `_predictions_within_regime_running_score_window` helper supports **rolling 6-hour naive-UTC** filtering for non-TFEX rows for analytics consistency, but **regime flatten / TFEX force-close behaviour executes only on derivatives sessions**.

Position sizing

Defaults per `TradeConfig` (each session may override percentages):

- **`high_*`** → **`high_conf`** of starting balance / risk budget (default **20%**)
- **`low_*`** → **`low_conf`** (default **5%**)
- Any other directional label text without **`high_`** / **`low_*`** prefix uses **`med_conf`** (default **10%**) — includes **`medium_*`**.

Executor-specific risk budget:

- **Bitkub paper:** `invest_pct * starting_balance`, converted through the limit rate.
- **TFEX:** `invest_pct * tfex_max_open_contracts` (whole contracts), executor-normalized.

TA emits **`units`** as **capacity hints** on opens (`max_buy_units` / `max_sell_units`); **closes** carry exact held sizes subject to caps.

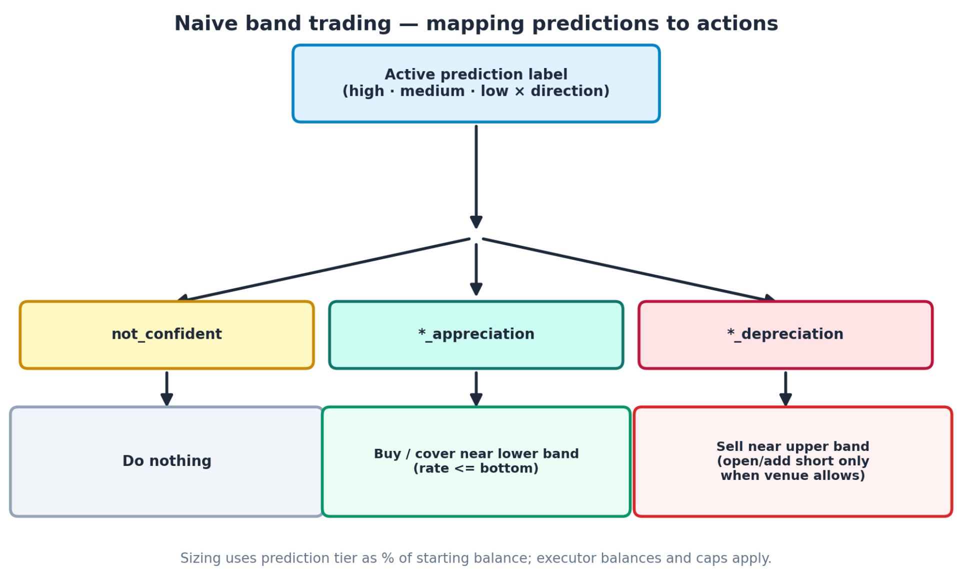

Figure 5: Overview flowchart for prediction labels versus naive band entries.

Entry sketch (combined behaviour)

- **Lower band, flat, balance > 0: buy** long — requires `appreciation` in the effective prediction text.

- **Lower band, short:** **buy** to cover — spot requires `appreciation`; TFEX covers on geometry once tick gate passes.
- **Upper band, long:** **sell** to flat — spot requires `depreciation`; TFEX closes on geometry.
- **Upper band, flat / long slot for shorts:** **sell** to open short — only if `short_ok` and text contains `depreciation`.
- **Upper band while net short:** upper-band **sell** path does **not** manufacture extra covers — messaging directs flatten via lower band / regime logic.

Short exposure

Bitkub spot / paper: long or flat only — sells without inventory are ignored.

TFEX: real shorts via Settrade derivatives executor; sizing, margin, and contract semantics are venue-owned.

Execution notes

- **Separation:** TA chooses **threshold geometry** and **limit references**; executors decide **filled size** from `invest_pct` and venue caps.
- **TFEX limits:** prices are **0.01** ticks (2 dp); executors typically apply a **one-tick aggressive nudge** on limits for fill reliability (`TradeActionSchema` defaults).
- **Lower-band buys:** limit basis uses `buy_rate` snapped to TFEX decimals when on derivatives.
- **Spot buy budget cap:** persisted paths clamp notionally against `min(total_balance, starting_balance)` where applicable so simulated spot books do not pyramid purely on profits.

Balance and PnL (high level)

- Quote cash moves by +/- `units * rate` on fills.
- Realized PnL follows average-cost long exits and average-entry short covers; opening shorts does not realize immediately.

Backtest dashboard

- Metrics emphasize **realized** PnL only (no unrealized MTM snapshot).
- **Net base units** come from the **last** trade row's `current_units` for that session.

Validation of prediction accuracy

Evaluation uses `session.rate_change_threshold` (default `0.01` rate units vs Bitkub closes when available):

- `|move| <= threshold` → counted correct only when the prediction text includes `no_change` (flat expectation).
- `move > threshold` → counted correct when the prediction includes `appreciation`.
- `move < -threshold` → counted correct when the prediction includes `depreciation`.

One-line summary

Forecasting (trigger → researcher → predictor, or schedules) persists labeled horizons; each **tick** runs `run_trading_pipeline`: **stop-loss**, then **mean-band TA (SMA +/- max(k * population_std, 0.01)** bands,

bid/ask touches, **spot-only label gates on closes**), then the **executor**, with TFEX-only tick/regime/force-flat extras — sized by `high_*` / `med_*` / `low_*` fractions of the session risk anchor.